

Schizophrenia is accompanied by changes in the size of neuronal nuclei and an increase in the frequency of long-range chromatin contacts in aging

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Motivation and Aim

Schizophrenia is a highly heritable psychiatric disorder, yet most risk-associated genetic variants identified by genome-wide association studies are located in non-coding regulatory regions rather than protein-coding genes. These findings suggest that altered gene regulation, rather than changes in protein structure, may contribute to disease pathogenesis. Because gene regulation is closely linked to three-dimensional genome organization, we investigated chromatin architecture in schizophrenia using Hi-C analysis of human postmortem brain tissue.

Results

Analysis of chromatin contact probability revealed altered age-dependent dynamics in neurons. Healthy donors exhibited an age related increase in long-range chromatin interactions, whereas this pattern was absent in schizophrenia. Fluorescence microscopy further demonstrated enlarged neuronal nuclei in young schizophrenia samples. Transcriptomic analysis revealed significant acceleration of molecular aging in schizophrenia compared with healthy controls.

Methods

Postmortem samples of the human cerebral cortex (Brodmann area 22) were used in this study. Nuclei were isolated from frozen tissue and separated into neuronal and glial populations by fluorescence-activated cell sorting. Hi-C libraries were generated and sequenced for both cell populations. Sequencing data were processed using the distiller-nf pipeline, and downstream analyses were performed in Python.

Results

Our findings indicate that schizophrenia is associated with accelerated transcriptomic aging and altered age-dependent remodeling of long-range chromatin architecture while preserving local chromatin organization. The overlap between aging- and schizophrenia-associated transcriptional programs, together with changes in higher-order genome organization and increased neuronal nuclear size, suggests that dysregulated genome architecture may contribute to accelerated brain aging in schizophrenia.

